Hybrid Wired and Wireless Network Design, Configuration and Simulation Using Cisco Packet Tracer

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1. Introduction

This growing dependence on digital communication systems has brought to the fore the significance of well-architected, scalable, and resilient computer networks in the contemporary organizational settings. A hybrid wired and wireless IPv4 network was to be designed, configured, and simulated using Cisco Packet Tracer in this coursework. The project entailed assigning a special block IP address that was based on the student ID, using classless addressing to create a number of subnets, setting up routing among three routers, configuring both wired and wireless equipment, and checking full device-to-device connectivity by using simulation and debugging tools (Nourildean et al., 2023). The last network involved wireless access, wired LAN segments, servers, routers, switches, and serial point-to-point links, which were combined into a logical and physical topology that was coherent.

This report is intended to provide a clear description of the design procedure, outline the actions that were undertaken to construct the network topology, chart the addressing and configuration procedures, and test the successful functioning of the network by carrying out a connectivity test.

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Subnet	Network Address	Mask	First Host Address	Last Host Address	Broadcast Address	Bit Mask
A (Wireless)	191.5.28.0	255.255.255.240 (/28)	191.5.28.1	191.5.28.14	191.5.28.15	11110000
B (Wired)	191.5.28.16	255.255.255.240 (/28)	191.5.28.17	191.5.28.30	191.5.28.31	11110000
C (Wired)	191.5.28.32	255.255.255.240 (/28)	191.5.28.33	191.5.28.46	191.5.28.47	11110000

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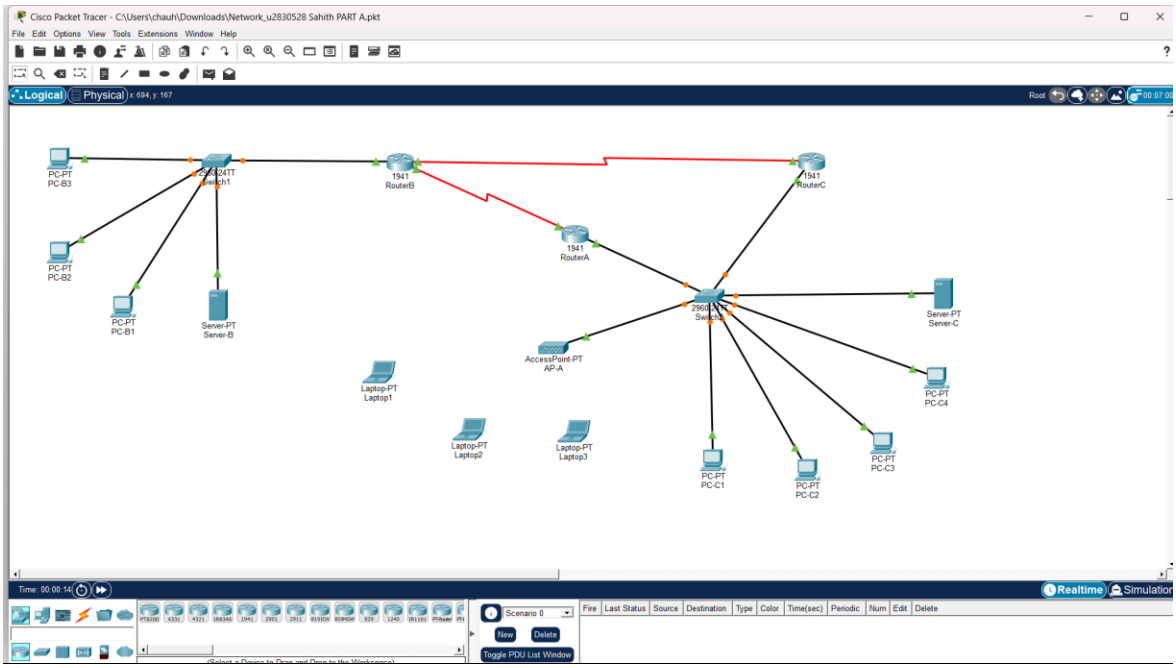

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Connection Point	Type of Cable / Medium Used
1. Between Routers and Switches	Copper Straight-Through
2. Between Routers	Serial DCE/DTE (Serial Cable)
3. Between Switches and Wireless Access Points	Copper Straight-Through
4. Between Switches and Computers	Copper Straight-Through
6. Between Switches and Servers	Copper Straight-Through
7. Between Wireless Devices and Access Point	Wireless Medium (802.11 Radio)

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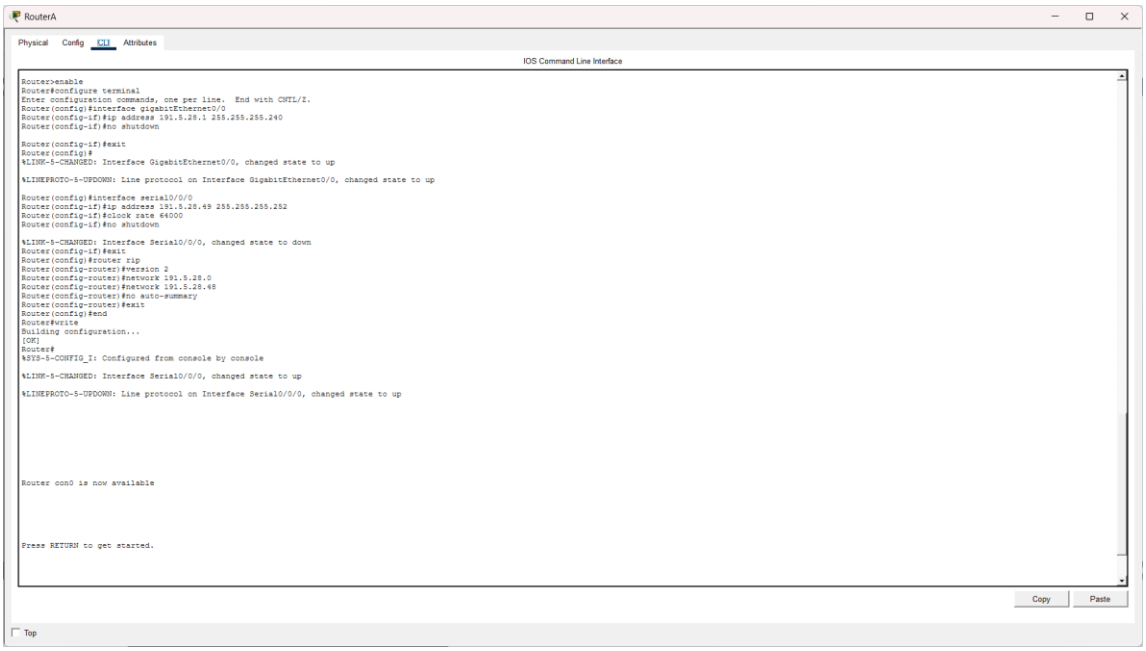

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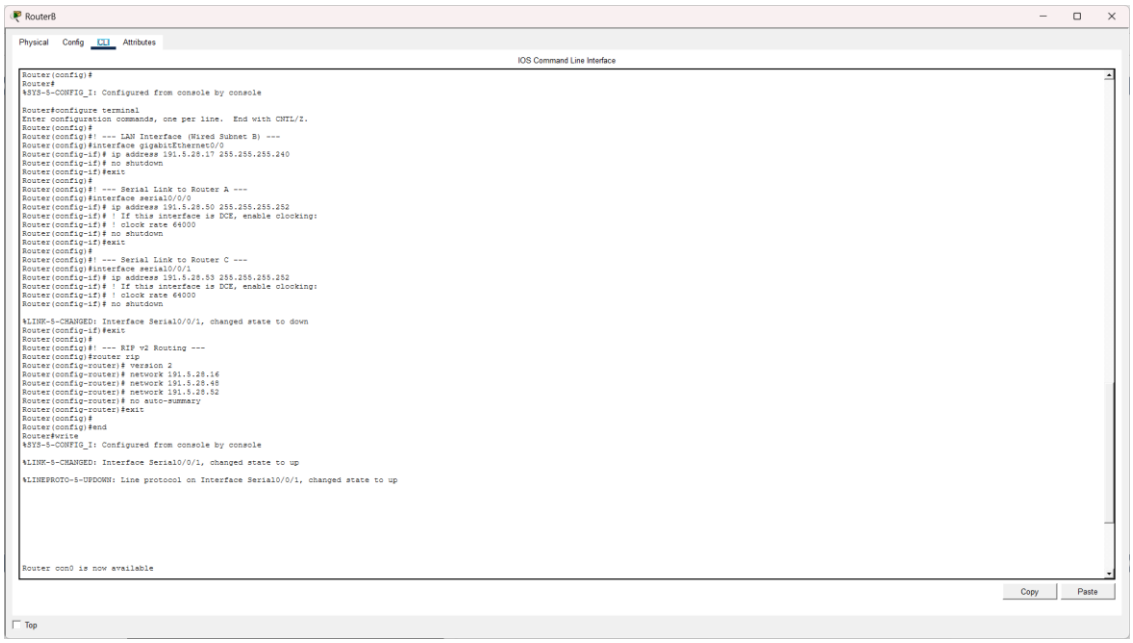

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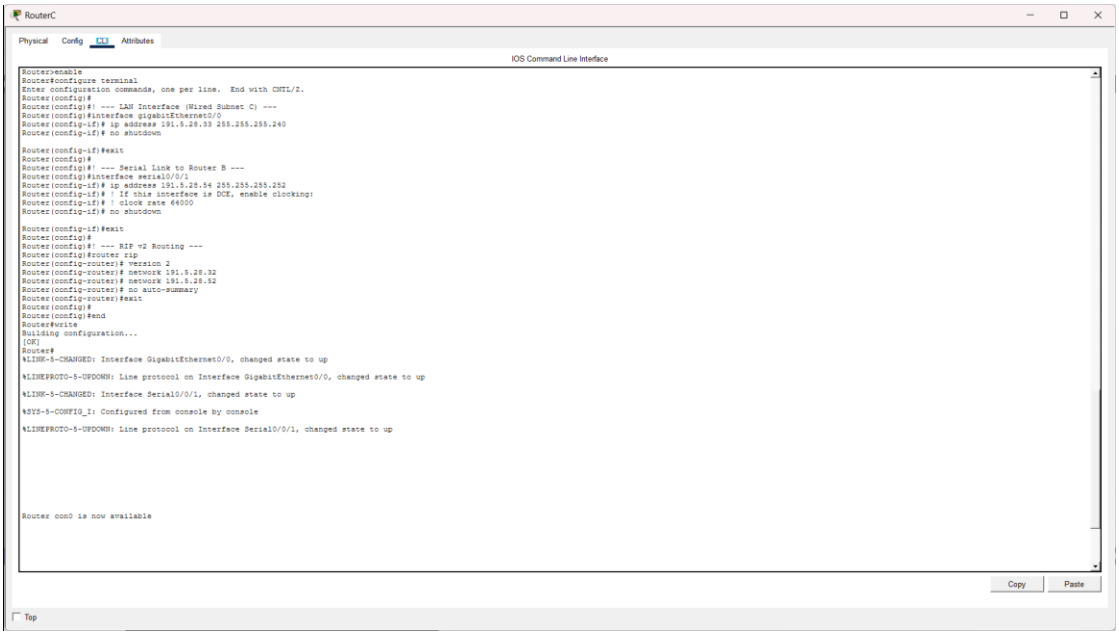

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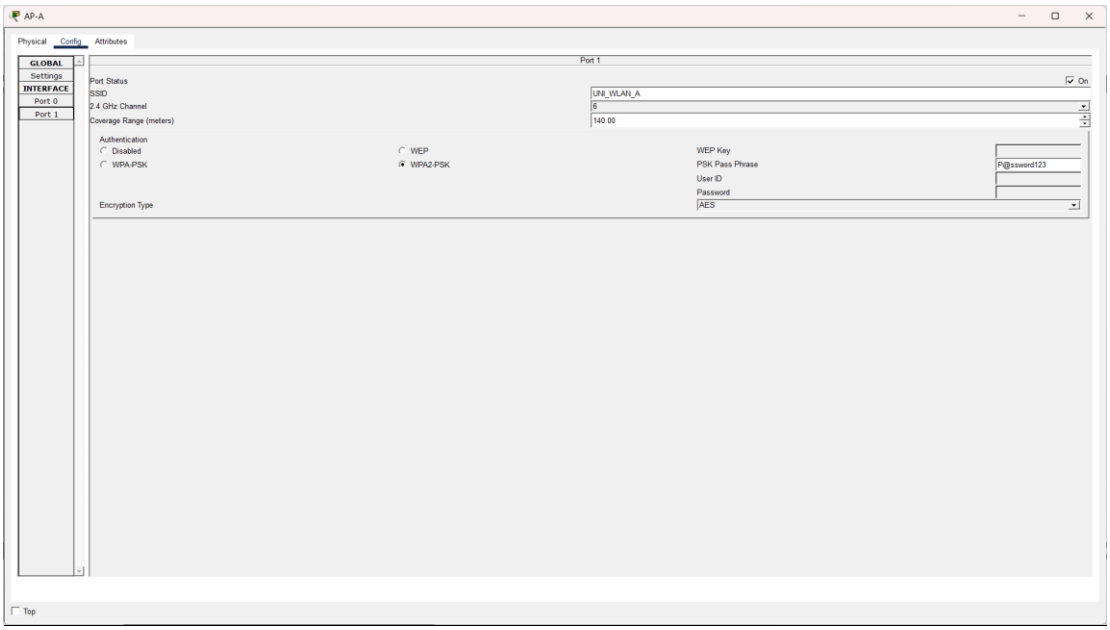

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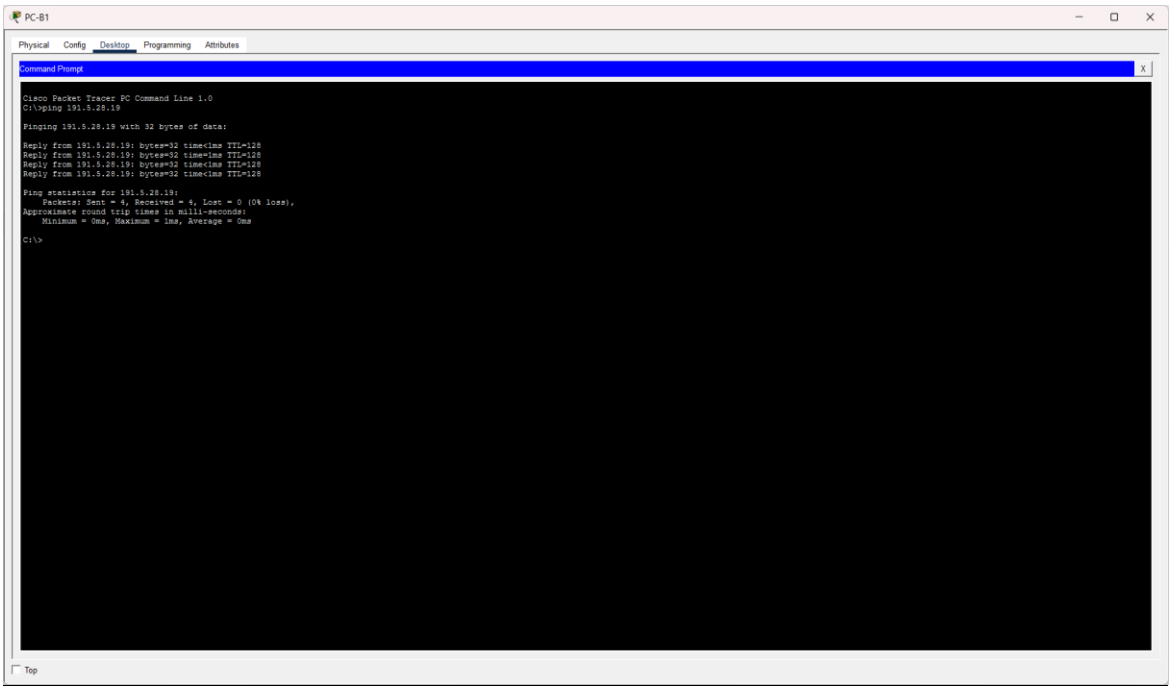

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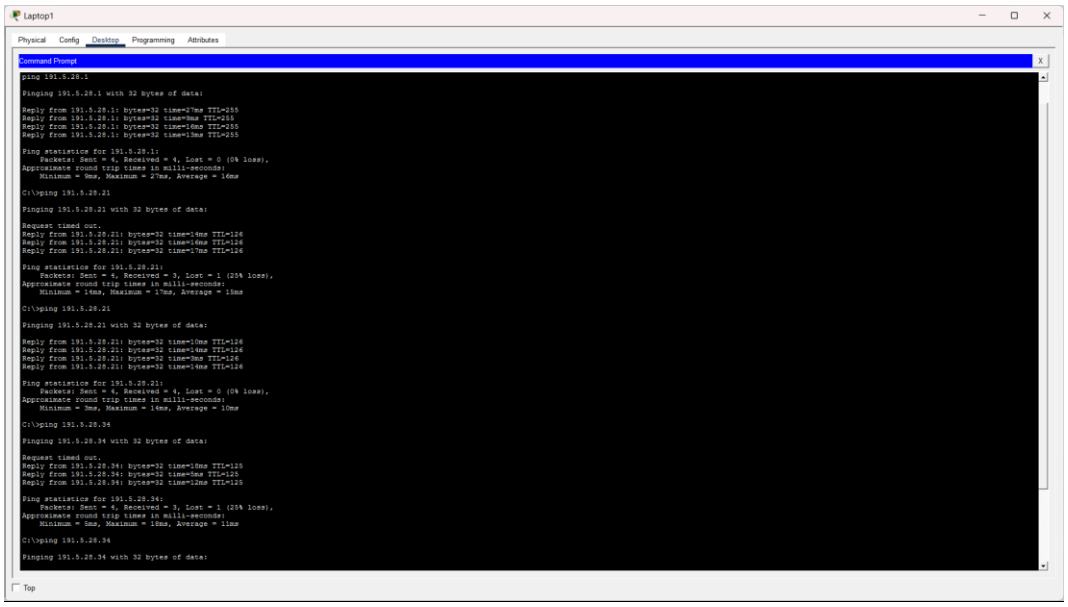

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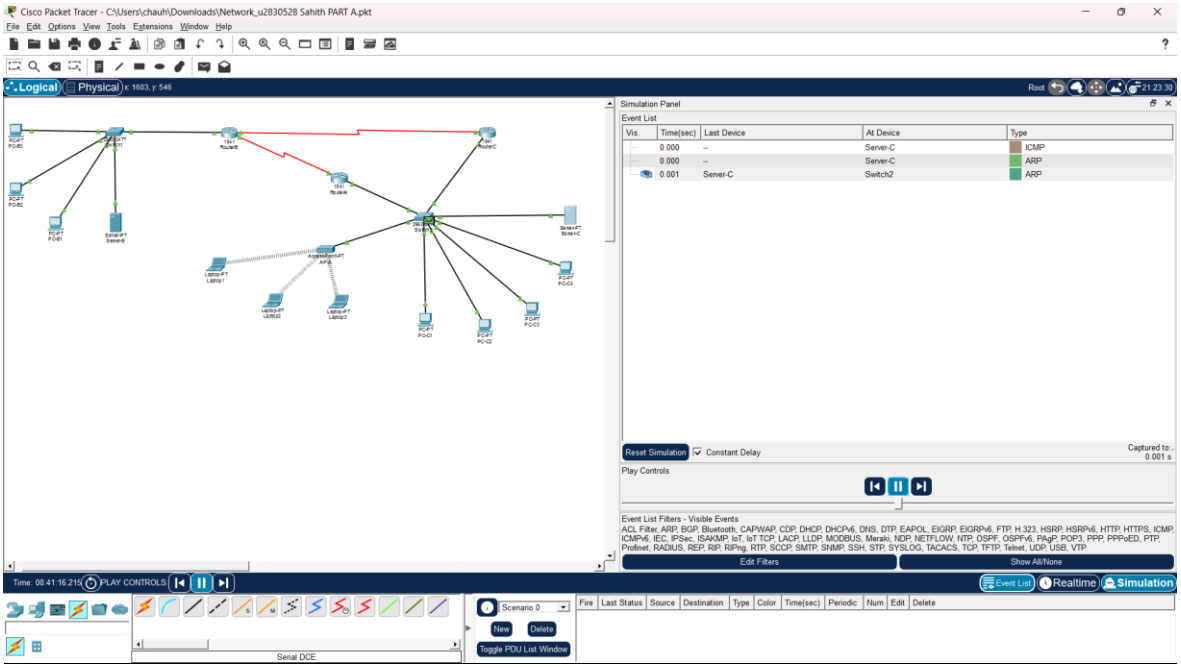
Cisco Packet Tracer - C:\Users\chauh\Downloads\Network_u2830528 Sahith PART A.pkt
File Edit Options View Tools Extensions Window Help
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Physical
Simulation Panel
Event List
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0.000 -- Server-C ARP
0.001 Server-C Switch2 ARP
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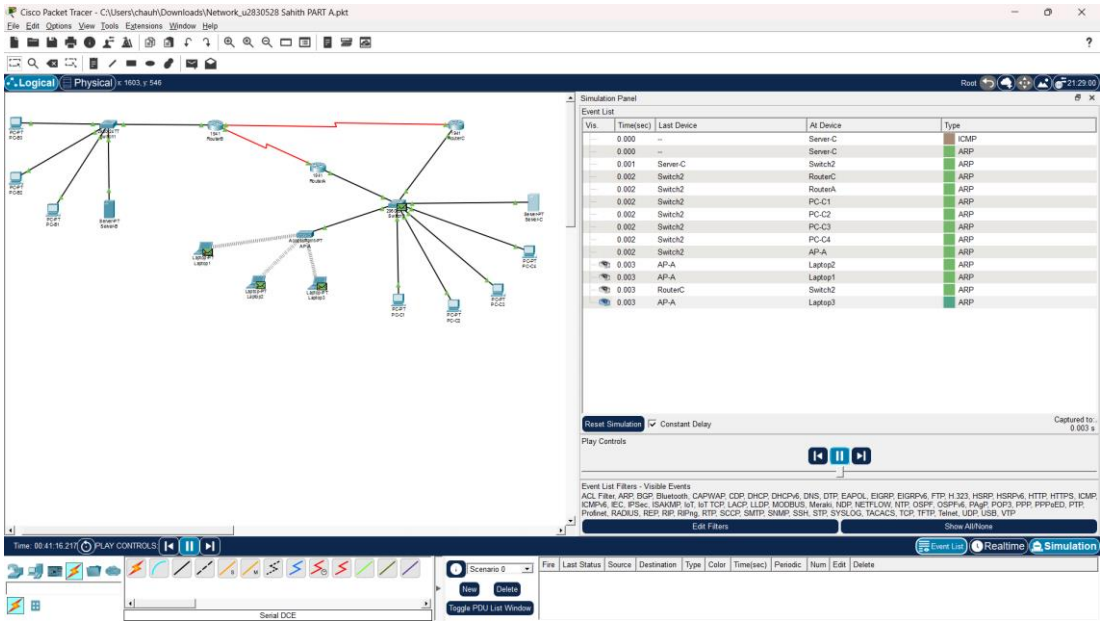

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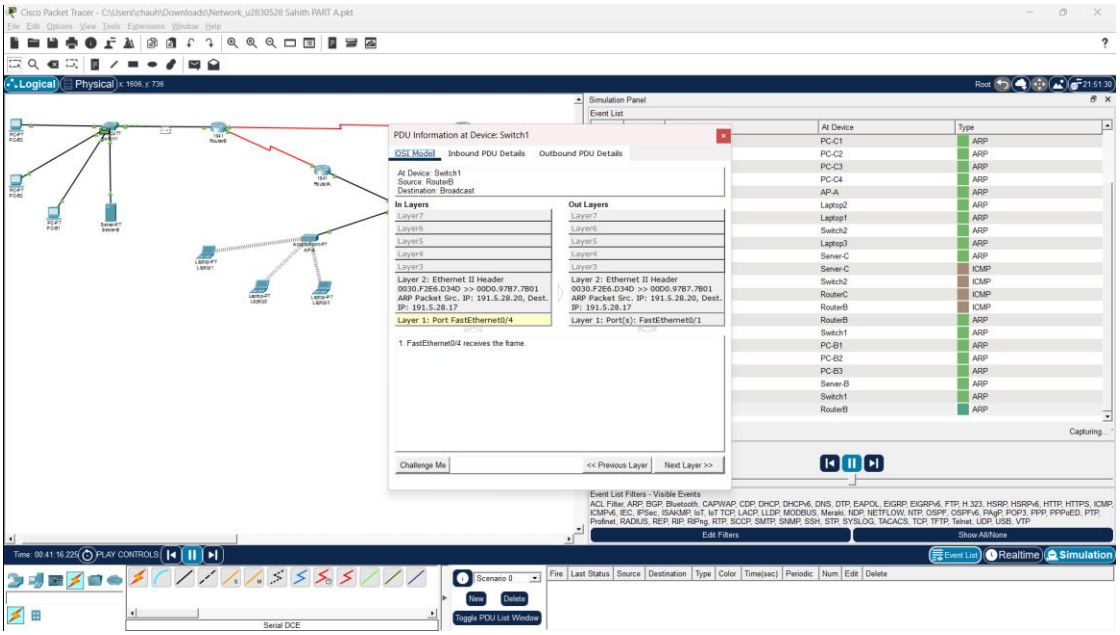

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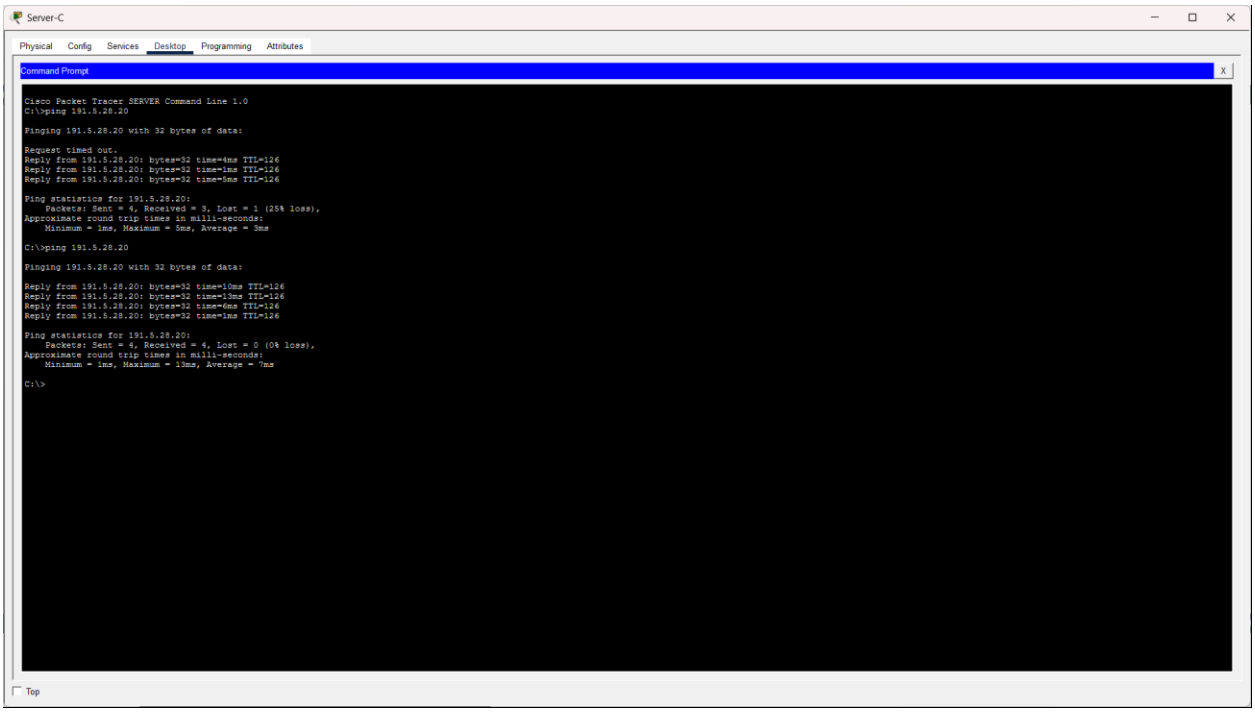

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Device	IP Address	IP Mask	Gateway Address
Host 1 (PC-B1)	191.5.28.18	255.255.255.240	191.5.28.17
Host 2 (PC-B2)	191.5.28.19	255.255.255.240	191.5.28.17
Host 3 (PC-B3)	191.5.28.20	255.255.255.240	191.5.28.17
Server (Server-B)	191.5.28.21	255.255.255.240	191.5.28.17

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Device	IP Address	IP Mask	Gateway Address	SSID	Security/Passkey
Wireless Access Point (AP-PT)	N/A	N/A	N/A	UNI_WLAN_A	WPA2-PSK / P@ssword123
Laptop 1	191.5.28.3	255.255.255.240	191.5.28.1	UNI_WLAN_A	WPA2-PSK / P@ssword123
Laptop 2	191.5.28.4	255.255.255.240	191.5.28.1	UNI_WLAN_A	WPA2-PSK / P@ssword123
Laptop 3	191.5.28.5	255.255.255.240	191.5.28.1	UNI_WLAN_A	WPA2-PSK / P@ssword123

Table: Connectivity Verification

From	To	IP Address	Results
Host 1 (B)	Host 2 (B)	191.5.28.19	S
Laptop 1 (A)	Gateway (Router A)	191.5.28.1	S
Laptop 1 (A)	Server (Subnet B)	191.5.28.21	S
Laptop 1 (A)	Host 1 (Subnet C)	191.5.28.34	S
Host 2 (B)	Laptop 2 (A)	191.5.28.4	S
Host 3 (C)	Server (C)	191.5.28.38	S
Server (C)	PC-B3	191.5.28.20	S
Router A	Router B Serial	191.5.28.49	S
Router B	Router C Serial	191.5.28.53	S